

School of Computer and Systems Sciences
Jawaharlal Nehru University
New Delhi
End-semester Examination
CS 406 - Numerical Methods
MCA 1st Semester
25th September 2025

Time: 2 hours

Total Marks: 20

Answer all questions

Q 1. What is linear independence? Explain how the concept of linear independence is important for solving a system of linear equations. 5

Q 2. Find the inverse of the coefficient matrix of the system of linear equations

$$x + y + z = 1$$

$$4x + 3y - z = 6$$

$$3x + 5y + 3z = 4$$

using the Gauss-Jordan method and then solve the system. 10

Q 3. Given the equation,

$$x^2 + 5 = e^x$$

first, find the interval in which the positive roots of the equation lie and then, using Newton-Raphson method, find the smallest positive roots of the equation. 5

School of Computer and Systems Sciences
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Mid-semester - 2 Examination
Numerical Methods (CS-406)
11th November 2025

Time: 2 hours

Total Marks: 20

Answer all questions.

Q 1. For the given the data below, find the polynomial $P(x)$ such that $P(x_i) = y_i \forall i = 0, 1, 2$ and then, find its first derivative, $P'(x)$ (8)

$$x_0 = 0, y_0 = 0$$

$$x_1 = 1, y_1 = 1$$

$$x_2 = 3, y_2 = 0$$

Q 2. Find the approximate value of


$$I = \int_0^1 f(x) dx$$

where

$$f(x) = \frac{1}{1+x}$$

using the trapezium rule with 4 equal subintervals. (8)

Q 3. What is the role of interpolation in animation. Explain in detail.
(4)

School of Computer and Systems Sciences
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New Delhi
CS - 406 Numerical Methods
End-semester Examination
5th December 2025

Time: 3 hours

Total Marks: 50

Answer all questions ($5 \times 10 = 50$ marks) .

Q 1. Find the positive roots of the equation $x^3 - 3x + 1 = 0$ using Newton Raphson method.

Q 2. Find the solution of the system of equations

$$\begin{aligned}x_1 + x_2 - x_3 &= 2 \\2x_1 + 3x_2 + 5x_3 &= -3 \\3x_1 + 2x_2 - 3x_3 &= 6\end{aligned}$$

using the Gauss-Jordan method.

Q 3. Using Lagrange's formula, find the interpolating polynomial that takes the following values.

$$x_0 = 0, y_0 = 0; x_1 = 1, y_1 = 1; x_2 = 3, y_2 = 0$$

Q 4. Derive the trapezium rule using the Lagrange linear interpolating polynomial. Find the approximate value of

$$I = \int_0^1 \frac{x}{x^3 + 10} dx$$

using the composite trapezoidal rule with 2 equal subintervals.

Q 5. Find the solution at $x = 0.4$ of the initial value problem using the fourth order Runge-Kutta method.

$$\frac{dy}{dx} = -2xy^2, y(0) = 1$$

Assume the step length $h = 0.2$.